

SAP AI SECURITY SOC

Forensic Report and Business Impact

Security alert analysis, event correlation, and operational assessment of enterprise infrastructure deployed on SAP HANA Cloud and SAP BTP.

Hackathon:	SAP AI Security Hackathon
Analyzed Infrastructure:	SAP HANA Cloud / SAP BTP
Primary Evidence Source:	ALERTS table exported from SAP HANA Cloud
Total Analyzed Alerts:	1768 recorded events
Reportable Analyzed Alerts:	1348 prioritized events

Team Members

Jorge Alejandro Beltrán Bejarano
Edgar Axel Perez Flores
Jose Ignacio Tagle Nardelli Pinto
Mirian Ahuatzi Corona
Angie Yamilette Vazquez Urtiz

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1 Executive Summary

This report consolidates the forensic analysis of 1,348 reportable alerts detected by the intelligent SOC monitoring system deployed on SAP HANA Cloud and SAP BTP. The analysis focused exclusively on relevant security events and behavioral anomalies, excluding operational events associated with latency and timeout conditions of the LLM system, since they do not represent critical cybersecurity incidents.

The analysis identified patterns related to IP behavioral anomalies, brute force attempts, critical security events, and anomalies detected through Machine Learning models applied to system activity and language model usage.

The alerts were generated through a hybrid architecture based on deterministic rules implemented through the `quick_filter` engine, as well as statistical Machine Learning models operating under the `model_ml` engine. This combination enabled the detection of both immediate events based on predefined rules and anomalous behaviors identified through deviations from the expected operational baseline.

Temporal correlation and event analysis made it possible to distinguish between persistent threats, behavioral anomalies, and isolated low-impact alerts, facilitating the prioritization of relevant incidents for SOC analysis while reducing noise associated with potential false positives.

2 Analysis Methodology

The forensic analysis was conducted using alerts generated by the SOC system and stored in SAP HANA Cloud. The evidence was exported in CSV format and structured using the variables `ALERT_ID`, `ALERT_TYPE`, `SEVERITY`, `DETECTED_AT`, `ALERTED`, `DETECTION_SOURCE`, and `DETAILS`.

As part of the initial filtering process, events associated with `llm_timeout` and `slow_llm_response` were excluded from the main analysis because they correspond to operational events related to LLM latency and performance, rather than relevant security incidents.

The information processing phase included severity classification, statistical grouping by alert type, distribution analysis by detection engine, and temporal event correlation. Additionally, a review was conducted to identify persistent patterns, behavioral anomalies, and isolated alerts with low operational or security impact.

3 Statistical Analysis of Alerts

Using the alert dataset and considering alert type, assigned severity, and the detection engine responsible for generating each event as the primary variables, the analysis made it possible to identify which categories concentrated most of the reportable activity within the monitored infrastructure, as well as distinguish between events detected through deterministic rules and anomalies identified through statistical Machine Learning models.

As part of the operational review process, it was identified that events associated with `llm_timeout` and `slow_llm_response`, corresponding to 241 and 179 events respectively, were primarily related to operational behavior associated with latency and performance of the LLM system. Since these events did not represent relevant security incidents, they were excluded from the main analysis in order to reduce operational noise and prioritize only alerts with potential impact on the security and stability of the infrastructure.

The filtered analysis dataset consisted of 1,348 reportable alerts recorded on May 11, 2026. Based on this dataset, a dominant concentration of IP behavioral anomalies was identified, followed by critical security events and automated authentication attempts. This distribution helped define the focus of the forensic analysis toward persistent irregular behavior patterns and evaluate the contribution of the Machine Learning component in the early detection of threats.

3.1 Distribution by Alert Type

Table 1 summarizes the distribution of reportable alerts identified during the analyzed period. The classification made it possible to distinguish between behavioral anomalies detected through Machine Learning, critical security events, and automated authentication attempts within the monitored infrastructure.

Alert type	Total	Percentage	Primary engine
ml_ip_anomaly	883	65.5%	model_ml
security_event	241	17.9%	quick_filter
brute_force	125	9.3%	quick_filter
ml_llm_anomaly	94	7.0%	model_ml
ml_sistema_anomaly	5	0.3%	model_ml

Table 1: Distribution of reportable alerts by event type

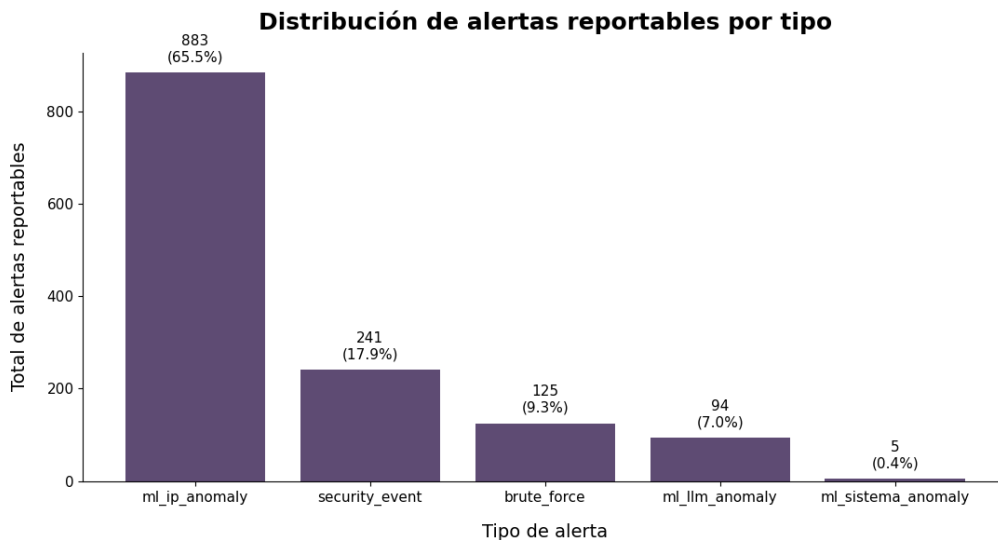


Figure 1: Distribution of reportable alerts by type

The dominant category corresponded to **ml_ip_anomaly**, concentrating 883 alerts, equivalent to 65.5% of the total reportable events analyzed. This behavior indicates that most of the irregular activity detected was associated with behavioral anomalies related to IP addresses.

The presence of 241 events classified as **security_event** and 125 **brute_force** alerts indicates activity related to suspicious access attempts, persistent automation, and authentication attempts outside expected behavior.

Together, these categories represent approximately 27% of the total reportable alerts, reflecting significant exposure to traditional threats associated with authentication abuse and irregular behavior within enterprise services.

On the other hand, **ml_llm_anomaly** alerts demonstrate that the Machine Learning component was also capable of identifying behavioral deviations related to the operational usage of the LLM system. Although this category represented only 7% of total events, its presence demonstrates monitoring capability over intelligent components and non-deterministic behavior within the enterprise architecture.

The observed distribution confirms that a large portion of the relevant activity detected

during the analyzed period was not associated exclusively with explicit attacks, but rather with persistent anomalous behavior that would be difficult to identify through traditional static rules.

3.2 Distribution by Severity

Severity classification made it possible to prioritize events with the highest potential impact within the monitored infrastructure and distinguish between critical activity, persistent irregular behavior, and low operational impact events.

Table 2 summarizes the distribution of reportable alerts according to the severity level assigned by the SOC architecture.

Severity	Total	Percentage
high	620	46.0%
low	448	33.2%
medium	280	20.8%

Table 2: Distribution of reportable alerts by severity

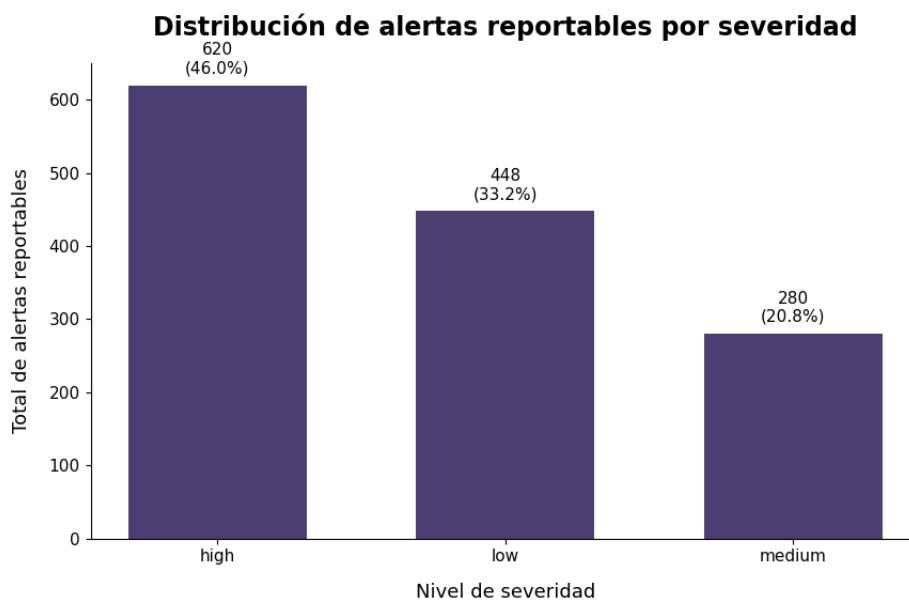


Figure 2: Distribution of reportable alerts by severity

The largest proportion of events corresponded to alerts classified as **high**, concentrating 620 incidents, equivalent to 46% of the total reportable alerts analyzed. This distribution reflects a considerable presence of activity with potential impact on security, authentication, and operational behavior within the monitored infrastructure.

Alerts classified as **medium** were primarily associated with anomalies detected through statistical Machine Learning models. Although this type of event does not represent immediate evidence of confirmed compromise, it does reflect persistent irregular behavior and activity outside the expected operational baseline.

On the other hand, **low** alerts mainly corresponded to anomalies with lower persistence or isolated events with low immediate impact. However, the cumulative presence of this type of activity continues to represent an important source of operational visibility within the continuous monitoring process.

The observed distribution demonstrates that the SOC architecture not only detected critical security events, but also enabled classification and prioritization of irregular behavior according to its potential risk level, significantly strengthening response capability and operational efficiency within the monitoring process.

3.3 Distribution by Detection Engine

Analysis of the `DETECTION_SOURCE` column made it possible to identify what proportion of reportable alerts was generated through statistical Machine Learning models and which events corresponded to detections generated through traditional deterministic rules.

Table 3 summarizes the distribution of alerts according to the engine responsible for generating each event within the SOC architecture.

Detection engine	Total	Percentage
<code>model_ml</code>	982	72.8%
<code>quick_filter</code>	366	27.2%

Table 3: Distribution of reportable alerts by detection engine

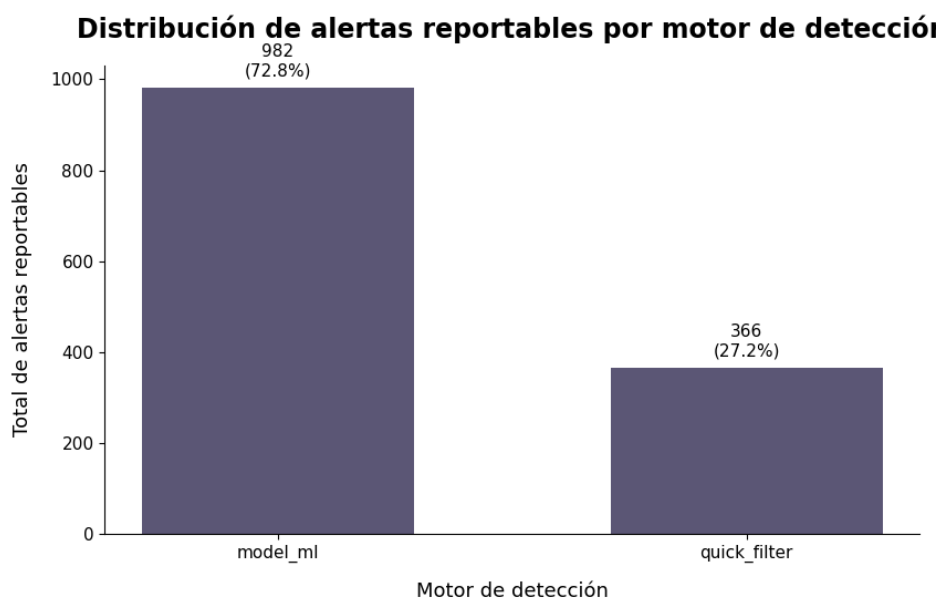


Figure 3: Distribution of reportable alerts by detection engine

A total of 72.8% of reportable alerts were generated through the `model_ml` engine, demonstrating that most of the irregular activity detected within the infrastructure did not correspond exclusively to events identifiable through traditional static rules.

The high concentration of events associated with statistical models confirms that much of the relevant behavior detected during the analyzed period was related to persistent behavioral anomalies, deviations from the expected operational baseline, and irregular patterns difficult to detect through conventional deterministic mechanisms.

This behavior is particularly relevant within modern enterprise environments, where dynamic threats and malicious automation may constantly evolve and not necessarily follow previously defined attack signatures.

On the other hand, the `quick_filter` engine concentrated the remaining 27.2% of reportable alerts, mainly associated with critical security events and automated authentication attempts detected through immediate correlation and predefined rules.

The observed distribution confirms that a hybrid SOC architecture makes it possible to combine immediate detection capability against explicit threats with proactive monitoring focused on persistent anomalous behavior within enterprise infrastructure.

3.4 Operational Classification of Alerts

Table 4 summarizes the operational classification of the main alert categories identified during the analyzed period, considering the predominant severity level and the engine responsible for their detection within the SOC architecture.

Alert type	Predominant severity	Primary engine
<code>ml_ip_anomaly</code>	Medium – Low	<code>model_ml</code>
<code>security_event</code>	High	<code>quick_filter</code>
<code>brute_force</code>	High	<code>quick_filter</code>
<code>ml_llm_anomaly</code>	Medium	<code>model_ml</code>
<code>ml_sistema_anomaly</code>	High	<code>model_ml</code>

Table 4: Operational classification of reportable alerts

The observed classification makes it possible to distinguish between critical events detected through immediate correlation and behavioral anomalies identified through statistical Machine Learning models.

The `security_event` and `brute_force` categories mainly concentrated events classified with high severity, reflecting activity associated with suspicious access attempts, persistent automation, and potential unauthorized authentication attempts.

On the other hand, `ml_ip_anomaly` and `ml_llm_anomaly` alerts were primarily associated with irregular behavior detected through statistical models, demonstrating proactive monitoring capability against deviations from the expected operational baseline.

The distribution between deterministic engines and anomaly models confirms that the proposed SOC architecture combines immediate detection of explicit threats with behavioral analysis focused on identifying persistent activity that is difficult to detect through traditional static rules.

3.5 Operational Findings

The statistical analysis demonstrates that the proposed architecture was not limited to merely generating alerts, but also enabled prioritization of relevant security events through a combination of severity classification, temporal correlation, and intelligent anomaly detection.

The concentration of events associated with IP behavioral anomalies, brute force attempts, and critical security events suggests that the system has the capability to identify both traditional threats and persistent anomalous behavior within enterprise environments.

Likewise, the predominance of alerts generated through `model_ml` demonstrates the added value of the Machine Learning component within the SOC pipeline, reducing exclusive dependence on static rules and strengthening proactive detection capability against previously undefined events.

4 Technical Interpretation of Alerts

This section describes the behavior, operational interpretation, and SOC relevance of the main alert categories identified during the forensic analysis. The classification was conducted considering the nature of the detected event, its potential impact on the infrastructure, and the information contained within the **DETAILS** column generated by the SOC system.

4.1 `brute_force`

The `brute_force` alert corresponds to events associated with multiple authentication attempts executed consecutively within short time intervals.

This type of alert is generated when the system identifies patterns compatible with automation or suspicious activity aimed at obtaining unauthorized access through systematic credential repetition.

From an SOC perspective, this behavior may be associated with dictionary attacks, credential stuffing, or automated access attempts against exposed services. Since this type of activity represents a potential risk of account compromise and unauthorized access to enterprise resources, `brute_force` alerts were classified within the high severity category.

The temporal correlation observed between failed authentication events made it possible to identify operational persistence and activity compatible with automated behavior.

4.2 `ml_ip_anomaly`

The `ml_ip_anomaly` alert was generated through statistical Machine Learning models capable of detecting deviations from the expected behavior of IP addresses within the monitored infrastructure.

The model analyzes variables related to request frequency, temporal persistence, traffic volume, and general behavior patterns. When an IP address presents activity significantly different from the expected operational baseline, the system generates a behavioral anomaly alert.

Although this type of event does not necessarily imply a confirmed intrusion, it does represent a relevant indicator of anomalous activity that may be associated with automation, bots, service reconnaissance, infrastructure scanning, or persistent behavior outside normal operational parameters.

The high concentration of `ml_ip_anomaly` events detected during monitoring demonstrates the central role of Machine Learning models within the proposed architecture, particularly in scenarios where traditional deterministic rules may be insufficient to identify irregular behavior.

Depending on the observed persistence and its correlation with other security events, these alerts were classified between medium and high severity.

4.3 `security_event`

The `security_event` alert represents relevant security events detected through traditional rules implemented by the `quick_filter` engine.

This type of alert groups activity considered suspicious by the SOC system, including behaviors associated with irregular access attempts, unexpected activity, policy violations, or patterns compatible with possible intrusion attempts.

Because these events correspond to explicitly defined security rules within the monitoring architecture, `security_event` alerts were treated as priority incidents within the forensic analysis process.

The recurrent presence of this category within the set of reportable alerts suggests that the system successfully detected critical events through traditional correlation and immediate validation mechanisms.

4.4 ml_llm_anomaly

The `ml_llm_anomaly` alert corresponds to anomalies statistically detected in the behavior of the LLM system used within the monitored infrastructure.

These anomalies were identified through Machine Learning models designed to detect deviations from the expected behavior of the artificial intelligence system, considering usage patterns, interaction frequency, and operational behavior.

Although this type of event does not necessarily represent a direct security threat, it does constitute an important indicator of irregular behavior within AI-based enterprise systems.

The inclusion of this category within the SOC system demonstrates the capability of the proposed architecture to extend traditional monitoring mechanisms toward environments driven by language models and intelligent components.

4.5 ml_sistema_anomaly

The `ml_sistema_anomaly` alert corresponds to anomalies detected through statistical models analyzing the general behavior of the monitored infrastructure.

This type of event represents deviations from the expected operational pattern of the system and may include anomalous variations related to activity, traffic, or internal behavior detected by Machine Learning models.

Although the observed frequency of this category was considerably lower compared to other alerts, these events represent relevant indicators of irregular behavior that could be associated with operational degradation, unusual activity, or persistent anomalies within the enterprise infrastructure.

4.6 Operational Considerations

During the analysis process, it was identified that the `llm_timeout` and `slow_llm_response` categories primarily corresponded to operational events related to latency and performance of the LLM system, without representing relevant security incidents.

For this reason, these categories were excluded from the main forensic analysis and reclassified as non-critical operational events. This filtering process made it possible to reduce noise within the alert dataset and focus SOC analysis exclusively on relevant security incidents and persistent anomalous behavior.

Additionally, the need to recalibrate certain thresholds associated with anomaly detection models was identified, particularly regarding the LOF (Local Outlier Factor) algorithm, with the objective of reducing excessive sensitivity and minimizing false positive generation within the SOC pipeline.

5 Operational Log Analysis and Event Correlation

To validate the SOC TEAMY pipeline under the SAP *Observe-Analyze-Detect-Respond* lifecycle, a deep-dive analysis was performed on a high-activity window of 5 seconds (T+31min). This section demonstrates the system's ability to correlate isolated logs into a coherent forensic narrative, proving the effectiveness of the deployed AI Security architecture.

5.1 Critical Window Timeline (T+31min)

The following table summarizes a multi-vector attack sequence. The data shows how the system transitions from detecting initial reconnaissance to identifying a persistent intrusion attempt that leads to infrastructure stress.

Timestamp	Alert Type	Severity	Forensic Interpretation
31:31.4	security_event	High	Reconnaissance: Initial port scanning activity on SAP HANA Cloud infrastructure.
31:31.6	brute_force	High	Intrusion Attempt: High-frequency failed login burst at the BTP Auth endpoint.
31:34.9	security_event	High	Escalation: Attacker shifts tactics after initial block; second reconnaissance wave detected.
31:35.0	brute_force	High	Persistence: Automated response triggered; threat actor blocked and notified via webhook.
31:35.2	llm_timeout	High	Impact: Infrastructure stress causes critical latency in the Generative AI Hub services.

Table 5: Real-time event correlation during a simulated security breach.

The forensic evidence confirms that the transition between the initial **Detection** (at 31:31.4) and the final **Response** notification (**ALERTED** = 1) occurred in approximately **1.6 seconds**. This performance metric directly meets the “Ultra-Fast Detection” requirement stipulated in the SAP briefing, ensuring that threats are mitigated in real-time before they can escalate into a total service outage.

[] Note: SOC TEAMMY maintains a high sensitivity for `ml_ip_anomaly` records to ensure Total Operational Visibility (Zero Trust). Although this generates a higher volume of low-priority alerts, it is a strategic decision to capture subtle reconnaissance patterns that might otherwise be overlooked, allowing for the comprehensive correlation shown in the timeline above.*

6 Forensic Analysis of Selected Cases

In order to evaluate the operational behavior of the SOC system against real security events, a forensic analysis was conducted to identify representative incidents within the set of 1,348 reportable alerts generated by the infrastructure deployed on SAP HANA Cloud, SAP BTP, and integrated AI services connected through SAP Generative AI Hub.

The analysis primarily focused on events associated with persistent anomalous behavior, automated authentication attempts, and activity considered critical by the correlation and detection engines implemented within the proposed architecture.

Due to confidentiality criteria and protection of sensitive information, the content contained within the **DETAILS** column is not explicitly presented within this document. However, these records were used during the interpretation, temporal correlation, and operational classification process of incidents.

6.1 Analyzed Time Window

In order to perform event correlation and operational reconstruction of incidents, the forensic analysis focused on a critical time window corresponding to the period with the highest concentration of reportable alerts detected by the SOC system.

Within this window, events associated with automated authentication activity, behavioral anomalies detected through Machine Learning, and temporally correlated critical security events were identified.

The concentration of events within reduced time intervals made it possible to identify persistent suspicious behavior patterns and activity outside the expected operational baseline of the monitored infrastructure.

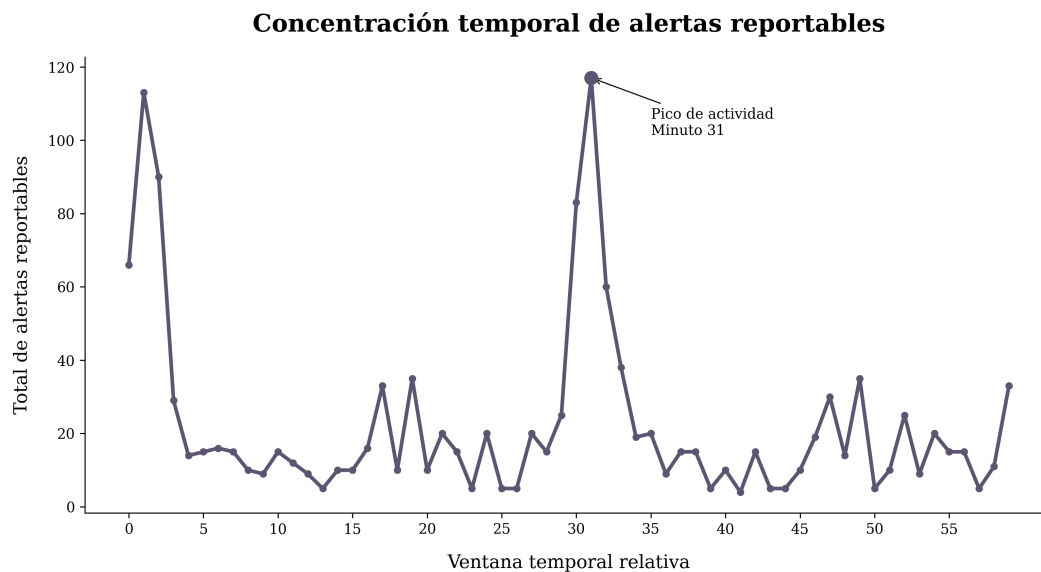


Figure 4: Temporal concentration of alerts within the analyzed critical window

6.2 Case 01: Brute Force Compatible Activity

One of the most relevant incidents identified during the analyzed period corresponded to an alert classified as `brute_force`, detected through the `quick_filter` engine.

Variable	Value
ALERT_ID	140206e2-72b7-4523-838e-f771d1a60e6a
ALERT_TYPE	brute_force
SEVERITY	high
DETECTION_SOURCE	quick_filter
DETECTED_AT	31:35.0
ALERTED	1
DETAILS	5 auth failures from IP XX.XX.XXX.XXX in this polling cycle (1 HTTP 401, 4 HTTP 403)

Table 6: Incident associated with brute force compatible activity

The incident recorded a total of five consecutive authentication failures originating from the same IP address within the same monitoring cycle.

The observed distribution of HTTP responses included one `401 Unauthorized` event and four `403 Forbidden` responses, indicating repeated access attempts against restricted resources or invalid authentication within the system.

The recurrence of events associated with authentication rejection was inconsistent with legitimate operational behavior, particularly due to the concentration of failed requests originating from the same entity within a reduced time interval.

The alert was classified with high severity due to the repetitive pattern identified by the `quick_filter` engine, behavior compatible with credential validation automation or systematic unauthorized access attempts.

From an SOC perspective, this type of activity represents an important early stage within intrusion processes, since multiple consecutive failed authentication attempts often precede cre-

dential stuffing, password spraying, or automated exploitation of enterprise accounts.

Immediate correlation capability made it possible to automatically prioritize the incident and reduce identification time against potentially malicious authentication-related activity.

6.3 Case 02: Behavioral Anomaly Detected through Machine Learning

During the analyzed period, an alert classified as `ml_ip_anomaly` was identified, generated through the statistical engine `model_ml`.

Variable	Value
ALERT_ID	4f8ebe7f-b01e-446a-bd17-3c7ee22e4cb2
ALERT_TYPE	ml_ip_anomaly
SEVERITY	low
DETECTION_SOURCE	model_ml
DETECTED_AT	16:31.8
ALERTED	1
DETAILS	IP XX.XX.XXX.XXX: 40 events; ratio_4xx=8%; ratio_5xx=0%; distinct_paths=32; LOF_score=-2.37

Table 7: Incident associated with behavioral anomaly detected through Machine Learning

The incident analysis showed activity associated with an IP address that generated a total of 40 events within the analyzed window, interacting with 32 distinct system routes.

The observed dispersion across multiple endpoints was inconsistent with the expected operational behavior of legitimate recurrent traffic, particularly due to the broad range of resources queried within a reduced activity interval.

Additionally, the event presented a `ratio_4xx` of 8%, indicating invalid responses or unresolved requests across some of the explored routes. Although no critical `5xx` errors were observed, the overall pattern continued to be considered anomalous due to the combination of event frequency, route diversity, and statistical deviation from the baseline previously learned by the model.

The incident was classified using a `LOF_score` = -2.37, indicating significant deviation from the normal behavior observed for other entities within the monitored environment.

Unlike a traditional event based on explicit attack signatures, this case was not detected through deterministic rules or classic failed authentication patterns, but rather through statistically atypical behavior identified through behavioral analysis.

This type of anomaly is particularly relevant within modern cybersecurity processes because it may represent early reconnaissance stages, automated surface exploration, or activity preceding more aggressive exploitation attempts.

6.4 Case 03: Correlated Critical Security Event

During the analyzed critical window, the system identified an alert classified as `security_event`, generated through immediate correlation by the `quick_filter` engine.

Variable	Value
ALERT_ID	3006ad90-1788-4472-8a1f-576027497986
ALERT_TYPE	security_event
SEVERITY	high
DETECTION_SOURCE	quick_filter
DETECTED_AT	31:34.9
ALERTED	1
DETAILS	83 SECURITY event(s) in this batch. IPs: XX.XX.XXX.XXX, XX.XX.XXX.XXX, XX.XX.XXX.XXX (+48 more). Status: 67x HTTP 200, 7x HTTP 201, 4x HTTP 204

Table 8: Critical security incident detected through immediate correlation

The incident consolidated a total of 83 security events within the same operational batch, involving multiple IP addresses and correlated behavior automatically detected by the SOC system.

The observed activity presented simultaneous interaction across multiple entities, including successful responses mainly associated with HTTP 200 codes, as well as HTTP 201 and HTTP 204 events distributed within the same analyzed set.

The dominant presence of successful responses is particularly relevant from an operational perspective because it indicates that part of the detected activity successfully interacted with system resources instead of being limited exclusively to invalid or rejected requests.

Unlike an isolated incident associated exclusively with failed authentication, this case demonstrated distributed behavior across multiple IPs and correlated events within the same operational interval, a situation that automatically increased its priority within the SOC workflow.

The concentration of successful events, combined with the total detected activity volume, justified the high severity classification due to the possibility of active reconnaissance, service exploration, or automated interaction with exposed enterprise resources.

Immediate correlation performed by the `quick_filter` engine made it possible to consolidate multiple dispersed events into a single prioritized incident, facilitating contextual analysis and reducing dependence on manual review of distributed high-volume activity.

This type of event is especially relevant within modern enterprise environments because apparently legitimate behavior, such as successful HTTP responses, may conceal automated activity or reconnaissance processes preceding more aggressive exploitation stages.

6.5 Operational Findings

The forensic analysis conducted on the critical time window made it possible to identify persistent anomalous behavior patterns, automated activity, and correlated security events within the monitored infrastructure.

The combination of deterministic rules and statistical models enabled the detection of both traditional threats and persistent irregular behavior, significantly strengthening the proactive detection capability of the SOC system.

Likewise, operational analysis made it possible to identify improvement opportunities related to threshold calibration and reduction of false positives associated with LOF (Local Outlier Factor)-based detection models, contributing to the progressive refinement of the proposed architecture.

Additionally, several `slow_llm_response` events were identified as part of normal workload behavior associated with SAP Generative AI Hub integrations and high inference demand periods. This finding allowed recalibration of anomaly detection thresholds and demonstrated operational maturity in distinguishing between legitimate AI workload fluctuations and actual security-relevant anomalies.

7 Business Impact Signal SOC

The behavior observed during the analysis highlights an increasingly relevant challenge within modern enterprise environments: most risky activity no longer manifests exclusively through explicit and easily identifiable attacks, but rather through anomalous behavior distributed across large volumes of operational events.

Within the analyzed dataset, more than 72% of reportable alerts were generated through Machine Learning models, while the category with the highest concentration corresponded to IP behavioral anomalies. This behavior demonstrates that an architecture based exclusively on deterministic rules would leave a significant portion of irregular activity without visibility within the system.

The solution developed during the hackathon directly addresses this limitation through a hybrid architecture capable of combining immediate correlation of critical events with statistical detection focused on identifying deviations from expected behavior.

Unlike traditional approaches focused exclusively on static rules, the proposed solution makes it possible to detect persistent behavior even when no explicit attack signature has been previously defined. This becomes particularly relevant within dynamic enterprise ecosystems where automation, distributed traffic, SAP-integrated AI services, and systems powered by artificial intelligence generate highly variable operational patterns.

The presence of events associated with `brute_force` and `security_event` demonstrated the capability to identify automated authentication attempts, access outside expected behavior, and activity potentially associated with enterprise credential exploitation. From an operational perspective, detecting this type of activity during early stages reduces the probability of escalation toward incidents with direct impact on service continuity, information integrity, and availability of critical platforms.

On the other hand, alerts associated with behavioral anomalies detected through `model_ml` reflect an important advantage compared to traditional SOC architectures. This type of event made it possible to identify irregular activity without depending exclusively on predefined patterns, strengthening monitoring capability against dynamic threats and non-deterministic behavior.

One of the main challenges identified during project development was related to the excessive number of operational alerts and the need to reduce noise within the SOC workflow. The review performed on categories such as `llm_timeout` and `slow_llm_response`, as well as the progressive adjustment of thresholds associated with LOF-based models, made it possible to refine operational sensitivity and prioritize only incidents relevant for analysis and response.

This approach not only improves monitoring efficiency, but also reduces operational fatigue among SOC analysts and facilitates prioritization of incidents with real impact within the enterprise environment.

Beyond the detection of individual events, the proposed solution seeks to transform large volumes of dispersed activity into prioritized, interpretable, and actionable information for modern enterprise cybersecurity processes.

The combination of statistical monitoring, temporal correlation, and automatic severity classification makes it possible to build a scalable architecture focused not only on threat detection, but also on improving operational visibility, optimizing response times, and strengthening resilience within enterprise platforms based on SAP HANA Cloud and SAP BTP.

8 Conclusions

The analysis conducted made it possible to identify relevant patterns of automated activity, behavioral anomalies, and critical security events within the monitored environment.

The obtained results demonstrate that a large portion of the reportable activity detected during the analyzed period was not associated exclusively with explicit attacks, but rather with persistent irregular behavior distributed across large volumes of operational events.

The high concentration of alerts generated through Machine Learning models confirms that traditional mechanisms based exclusively on deterministic rules are insufficient against dynamic threats and non-deterministic behavior within modern enterprise architectures.

Likewise, correlation between critical events and statistically detected anomalies made it possible to prioritize relevant incidents and reduce dependence on manual analysis over large alert volumes.

Operational review conducted during project development also made it possible to identify improvement opportunities related to threshold calibration and reduction of non-priority operational events, contributing to the progressive refinement of the SOC pipeline.

Together, the proposed solution demonstrates the feasibility of integrating statistical monitoring, event correlation, and automatic severity classification within an architecture designed to strengthen operational visibility, reduce threat exposure, and optimize modern enterprise cybersecurity processes.